

Economics, Law & Ethics

The 80/20 Revision Guide

Part IB — Cambridge CST — Paper 7

How this guide is organised

ELE sets two questions a year on **Paper 7**, usually one economics-leaning and one law/ethics-leaning. Across **eight years (2018–2025, 15 questions)** the frequencies are unusually *flat* — this is a breadth course:

Topic	Appearances	Share
Game theory & (micro)economics	4	27%
Information asymmetry & market failure	3	20%
Economic models of technology markets	3	20%
Legal frameworks for data & technology	3	20%
GDPR & data protection	3	20%
Philosophical approaches to ethics	3	20%
Hacker ethics & cybercrime law	3	20%
Automated decision-making / AI ethics & regulation	4	27% (combined)
Intellectual property; privacy & surveillance	2 each	13%
Auctions; monopolies; AI consequences	1 each	7%

80/20 verdict. There is **no single dominant topic** — the exam rewards *breadth across three pillars*. **Economics** and **Ethics** are the heaviest, with **Law** close behind and the most *factually precise* (memorise the statutes). The single most-examined topic is **game theory**, and **AI ethics / automated decision-making** is now the fastest-growing theme. Strategy: secure *one strong answer in each pillar* — you only need two questions, so depth in two pillars plus a safety net in the third is enough. Lecturer's angle (Hutchings/Clayton): a security-economics and cybercrime slant, hence the prominence of *hacker ethics* and the *Computer Misuse Act*.

1 Economics (heaviest pillar)

1.1 Game theory & auctions (the top single topic — 27%)

Exam-critical

Nash equilibrium: a strategy profile where no player can do better by *unilaterally* changing strategy. A **dominant strategy** is best regardless of others' choices.

Prisoner's dilemma: mutual defection is the unique Nash equilibrium yet is *Pareto-dominated* by mutual cooperation — individually rational, collectively bad. Models security under-investment, spam, doping, etc.

	Cooperate	Defect
Cooperate	(3, 3)	(0, 5)
Defect	(5, 0)	(1, 1)

Iterated games. With repetition (unknown horizon), cooperation can be sustained — *tit-for-tat* (Axelrod) is nice, retaliatory, forgiving, clear. A known final round unravels by backward induction.

Worth knowing

Auctions. Four canonical types: *English* (ascending open), *Dutch* (descending open), *first-price sealed-bid*, *second-price sealed-bid* (*Vickrey*).

- **Vickrey is truthful** (incentive-compatible): bidding your true value is a dominant strategy, because what you *pay* is set by someone else's bid.
- **Revenue-equivalence theorem:** under independent private values + risk-neutral bidders, all four yield the *same expected revenue* to the seller.
- **Winner's curse:** in *common-value* auctions the winner is the one who most over-estimated — rational bidders *shade* their bids down.
- **Mechanism design** (“reverse game theory”) engineers the rules so that self-interested play yields the desired outcome (e.g. spectrum auctions).

1.2 Market failure & asymmetric information (20%)

Exam-critical

A **market failure** is an allocation where the free market is not Pareto-efficient. The CS-relevant kinds:

- **Externalities** — costs/benefits fall on third parties (pollution; insecure machines → bot-nets). Remedy: tax/subsidy (Pigouvian), regulation, assign property rights (Coase).
- **Public goods** — non-rival & non-excludable (open-source, security research) → free-riding, under-provision.
- **Tragedy of the commons** — shared resource overused because each user bears only part of the cost (IPv4 space, bandwidth, attention).
- **Monopoly** — price above marginal cost; deadweight loss; monopoly rents.

Worth knowing

Asymmetric information (one side knows more):

- **Adverse selection** (*hidden information, before the deal*) — Akerlof's “market for lemons”: bad products drive out good because buyers can't tell them apart. Fixes: *signalling* (warranties, certificates), *screening*.
- **Moral hazard** (*hidden action, after the deal*) — insured parties take more risk. Fixes: deductibles, monitoring, aligned incentives.

Security economics (Anderson): systems often fail because the party best placed to fix a problem is not the party who bears the cost — *misaligned incentives*, not bad crypto.

1.3 Economics of technology markets (20%)

Worth knowing

Information goods have **high fixed / near-zero marginal cost** (Shapiro & Varian, *Information Rules*), so pricing is value-based, not cost-based.

- **Network effects** — a good's value rises with the number of users; **Metcalf's law**: value $\propto n^2$. Drives *positive feedback* and *tipping* (winner-take-all, standards wars).

- **Lock-in & switching costs** — proprietary formats, learning, data; total cost includes the cost to switch. Vendors engineer lock-in.
- **Price discrimination** — charging by willingness-to-pay: 1st degree (personalised), 2nd (*versioning*, self-selection), 3rd (group/student pricing). **Bundling** (e.g. Office) extracts more surplus by smoothing valuations.
- **Two-sided markets / discriminating monopolist / dominant-firm** models complete the toolkit.

2 Law (most factually precise pillar)

Gotcha

Law answers are marked on **naming the right statute and case**. “There’s a law against it” scores nothing; “s1 Computer Misuse Act 1990, as in *R v Cuthbert*” scores. Learn the Acts and one case each. (And remember: *IANAL* — this is UK law.)

2.1 Data protection: GDPR → DPA 2018 → DUAA 2025 (20%)

Exam-critical

GDPR (EU, in force 2018-05-25), implemented in the UK as the **Data Protection Act 2018**, now amended by the **Data (Use and Access) Act 2025 (DUAA)**. Protects the **data subject**. **Six principles**: data must be (1) fairly & lawfully processed; (2) for limited, specified purposes; (3) adequate, relevant, not excessive; (4) accurate & up to date; (5) not kept longer than necessary; (6) processed securely. Extra protection for *sensitive* personal data.

Worth knowing

- **Controller vs processor**: the controller decides *how/why*; the processor acts on its behalf and has its own obligations.
- **Lawful bases**: *consent* (freely given, specific, informed, unambiguous — no pre-ticked boxes), contract, legal compliance, vital interest, public interest, legitimate interest. Child under 13 needs parental consent.
- **Data-subject rights**: be informed, access, rectification, *erasure* (“right to be forgotten”), restrict, *portability*, object, and rights over *automated decision-making / profiling*.
- **Obligations**: data protection *by design & by default*, DPIAs, breach notification **within 72 hours**, appoint a *Data Protection Officer* (at-scale/public bodies). **Fines up to €20m or 4% of global turnover** (BA → £20m; Marriott → £18.4m).

2.2 Computer Misuse Act 1990 & cybercrime (the “hacker ethics” tag — 20%)

Exam-critical

Computer Misuse Act 1990 (prompted by *R v Gold & Schifreen*, where Prestel hacking convictions under forgery law were overturned):

- **s1** — unauthorised access (must *know* it’s unauthorised; need not be a specific machine or even in the UK).

- **s2** — s1 *with intent* to commit a further offence (up to 5 years).
- **s3** — unauthorised modification (up to 10 years); written to criminalise *virus-writing*, amended 2008 to cover *denial of service* and to make making/distributing *hacking tools* an offence (dual-use → hinges on *intent* / “likely to be used”).

Worth knowing

The test of “unauthorised”: “*if I were to ask, would they say yes?*” (*R v Lennon*, 2005 — mail-bombing held a s3 offence). Other cases: *R v Pile* (“Black Baron”, custodial for viruses); *R v Bedworth* (“addiction” defence, acquitted — couldn’t form intent); *R v Cuthbert* (the “tsunami hacker”, s1 for probing .././ URLs); *Allison/AMEX* (multi-level access can matter). The **Council of Europe / Budapest Convention** on cybercrime is the international framing.

2.3 Other law you should be able to name (20% “legal frameworks” + IP)

Worth knowing

- **Computer evidence:** *Civil Evidence Act 1968* (computer records admissible, not hearsay); *PACE 1984 s69* once required expert proof a machine worked — repealed (Youth Justice & Criminal Evidence Act 1999), so courts now *presume* correct operation unless disputed.
- **Interception:** *Investigatory Powers Act 2016* (replaced RIPA 2000, post-Snowden) — interception needs a **warrant signed by the Secretary of State**; intercept product is *not* admissible in court; exceptions for both-party consent and *lawful business practice* (must warn users). *RIPA Part III* can demand *decryption keys* (forgetting a key is a defence, but prosecution must prove you remember). Driven by *Malone* (ECtHR) and Art 8 ECHR (privacy).
- **Intellectual property:** copyright (automatic, software as a literary work), patents, trademarks, trade secrets, database right. (13%)
- **E-commerce:** Consumer Contracts Regs (identify seller, right to cancel); E-Commerce Directive (country-of-origin; *ISP immunities*: hosting, caching, mere conduit); Rome II (which law), Brussels Regulation (which court).
- **PECR 2003:** bans unsolicited marketing email to *natural persons* (soft opt-in); *cookies* need consent unless “strictly necessary”.

3 Ethics (joint-heaviest pillar)

3.1 The philosophical frameworks (20% — state them, then apply)

Exam-critical

- **Consequentialism / utilitarianism** (Bentham, Mill) — judge acts by *outcomes*; greatest good for the greatest number. *Act vs rule* utilitarianism. Weakness: can justify harming a minority.
- **Deontology** (Kant) — judge by *duties/rules*; the *categorical imperative* (act only on a maxim you could will universal; treat people as *ends*, never merely as means). Weakness: rigid, conflicting duties.
- **Virtue ethics** (Aristotle) — judge *character*; what would a virtuous agent do?

- **Rawlsian justice** — principles chosen behind a *veil of ignorance* (original position); the *difference principle* (inequalities only if they benefit the worst-off).

Worth knowing

Professional & research ethics: the **ACM** and **BCS** codes of conduct (public interest first, competence, honesty); research ethics requires *informed consent*, minimising harm, ethics-committee approval (cf. the Menlo/Belmont principles). Relevant to the Part II project evaluation.

3.2 Applied: AI ethics, privacy & the hacker ethic (the growth area — 27%+)

Worth knowing

- **Algorithmic bias & automated decisions:** training data encodes historical bias (predictive policing, hiring, credit). Ties to GDPR rights over automated decision-making (a right to *meaningful information* about the logic). Themes: fairness, accountability, transparency, *contestability*, the EU AI Act's risk tiers.
- **Privacy & surveillance:** mass vs targeted; the *nothing-to-hide* fallacy; chilling effects; data minimisation as an ethical *and* legal principle; the panopticon.
- **The hacker ethic** (Levy): information wants to be free; mistrust authority; the hands-on imperative. Frames the **vulnerability-disclosure** debate — *full* vs *responsible/coordinated* disclosure vs non-disclosure — as a trade-off between pressuring vendors and arming attackers.
- **Other contemporary issues:** censorship, targeted advertising, environmental impact (energy of large models), the ethics of LLMs.

Gotcha

Ethics answers fail when they're just opinion. **Structure:** (1) state the issue and stakeholders; (2) analyse it through *at least two named frameworks* (e.g. a utilitarian *and* a deontological reading), showing they can disagree; (3) reach a reasoned conclusion. Connect to law where relevant (GDPR, CMA) and to economics (externalities, misaligned incentives) — cross-pillar links score well in a course explicitly about their intersection.

Exam checklist

Exam-critical

1. **Game theory:** Nash equilibrium, dominant strategy, prisoner's dilemma (defection is Nash but Pareto-dominated), iterated games / tit-for-tat. Auctions: four types, *Vickrey truthful*, revenue equivalence, winner's curse.
2. **Market failure:** externalities, public goods, tragedy of the commons, monopoly; *adverse selection* (lemons, before) vs *moral hazard* (after); misaligned-incentive security economics.
3. **Tech markets:** network effects + Metcalfe ($\propto n^2$), lock-in/switching costs, price discrimination (1st/2nd/3rd, versioning), bundling.
4. **Data protection:** GDPR $\rightarrow$ DPA 2018 $\rightarrow$ DUAA 2025; six principles; controller vs processor; lawful bases; subject rights (incl. erasure, portability, automated decisions); 72-hour

breach; fines (€20m / 4%).

5. **Cybercrime:** CMA 1990 s1/s2/s3 (+ 2008 amendments: DoS, tools); “if I asked, would they say yes?”; cite a case (Gold & Schifreen, Lennon, Cuthbert). Computer evidence (PACE s69 repealed → presumption); IPA 2016 interception (SoS warrant, not admissible); RIPA Part III key disclosure.
6. **Ethics:** define consequentialism / deontology (categorical imperative) / virtue / Rawls; ACM/BCS codes + research consent. Apply *two* frameworks to an AI/privacy/disclosure case; link to law & economics. Don’t just give opinion.